

HIMA SRI YALAMANCHILI

AI/ML Engineer — Fresher | RAG Systems · LLM Applications · Full-Stack AI

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PROFESSIONAL SUMMARY

AI Engineering Intern at Calibo, currently building a **production AI-integrated triage backend** for government Primary Health Centers — combining **ML classification, FAISS vector search, and deterministic rule-based escalation** into a scalable FastAPI system. IBM-certified in **Retrieval-Augmented Generation**; built **6+ end-to-end AI/ML projects** spanning RAG, computer vision, and full-stack systems, all live and publicly accessible. Ranked in a national hackathon (**India RUNS 2026, Redrob AI × Hack2skill**) for a 3-layer hybrid candidate-ranking engine. CGPA **8.3/10**. Seeking an AI/ML Engineer role to build production-grade GenAI systems.

TECHNICAL SKILLS

AI/ML: LangChain, FAISS, HuggingFace Transformers, Sentence-Transformers, Gemini API, Qwen2.5-7B, OpenCV, Scikit-learn, NumPy, Pandas

Languages: Python, Java, C, JavaScript

Backend & Web: FastAPI, Flask, React, Tailwind CSS, SQLAlchemy, JWT Auth, REST APIs, SQLite, PostgreSQL

Deployment/Tools: Docker, Kubernetes (basic), Git/GitHub, Vercel, Render, Streamlit Cloud, HuggingFace Spaces

KEY PROJECTS

Outpatient Symptom Triage Assistant — Bilingual Healthcare AI (Academic Foundation) *(Python, FastAPI, PostgreSQL, Docker, Kubernetes)*

- Architected a **bilingual Telugu/English triage chatbot** for rural PHCs with **P1/P2/P3 urgency classification** and full **source-traceability** documentation — the academic prototype later extended into the production Calibo internship above.

Hybrid AI Candidate Ranking System — India RUNS Hackathon 2026 *(Python, FAISS, Sentence-Transformers)*

- Designed a **3-layer hybrid scoring engine** — semantic embeddings (**FAISS + Sentence-Transformers**), rule-based scoring, and a behavioral availability multiplier — validated against **4 adversarial test cases**.

AI Study Dashboard — RAG Learning Platform *(Python, Flask, LangChain, FAISS, Gemini, React, Vercel)*

- Built a **RAG pipeline** restricting the LLM strictly to uploaded curriculum PDFs, eliminating hallucinations on out-of-scope queries.
- Achieved **sub-2-second query response** with zero-cost local vector search (**FAISS + HuggingFace embeddings**); shipped full-stack with React + JWT role-based access.

Next-Gen Job Portal — Full-Stack SaaS *(Flask, SQLAlchemy, Render.com)*

- Production-deployed a job portal with **3 role-based dashboards** and real-time applicant tracking; secured with Flask-Login + Werkzeug hashing.

Additional Projects: Presentation LLM — Qwen2.5-7B-powered .pptx generator with 8 auto-selected color themes (**Streamlit, HuggingFace**) | Air Canvas — real-time hand-gesture drawing app at **25–30 FPS** using OpenCV contour detection.

EXPERIENCE

AI Engineering Intern | Calibo · Remote Apr 2026 – Present

- Designed and built an **AI-integrated backend service** for outpatient triage at government PHCs, combining **ML classification, FAISS vector search, and rule-based escalation** into a scalable production system.
- Engineered a **high-throughput FAISS similarity search service** over high-dimensional symptom vectors, enabling **low-latency case retrieval** and real-time intelligent ranking within the triage pipeline.
- Implemented a **deterministic rule engine** for emergency escalation that intercepts requests **before AI scoring**, enforcing reliability guarantees for critical cases independent of model behavior.
- Developed and deployed **RESTful FastAPI prediction endpoints** with structured request validation and error handling, delivering a backend built for large-scale PHC production deployment.

Data Analytics Intern (Python/Pandas) — Growth Track | Infinity Startup · Remote Aug – Nov 2025

- Analyzed campaign data across 3 platforms using **Python (Pandas)** to surface engagement trends, informing a strategy pivot that raised average post reach by **~18%**.
- Automated weekly KPI dashboards, cutting reporting time from **3 hours to under 30 minutes**.

EDUCATION

B.Tech, Information Technology — Dhanekula Institute of Engineering and Technology (DIET), Vijayawada

CGPA: **8.3/10** | Expected May 2027 | Relevant Coursework: Data Structures, DBMS, Web Technologies, OOP, Software Testing

CERTIFICATIONS & ACHIEVEMENTS

- IBM — Retrieval-Augmented Generation (RAG)** | IBM SkillsBuild — Emerging Technologies | Google — Gemini AI & Cloud Cybersecurity | Microsoft Launchpad — Cloud Computing | NPTEL — Blockchain & Human Interaction | Infosys Springboard — Python
- India RUNS Hackathon 2026** (Redrob AI × Hack2skill) — built candidate-ranking system + Ideathon growth-strategy pitch.
- 286+ LeetCode problems solved** (arrays, trees, graphs, DP); top 30% in weekly contests. 6+ AI/ML and full-stack projects deployed and publicly live.